北京交通大學

信号与系统

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and Information Engineering

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Frequency domain analysis of signals

- Frequency domain analysis of continuous periodic signals
- Frequency domain analysis of continuous non-periodic signals
- Frequency domain analysis of discrete periodic signals
- Frequency domain analysis of discrete non-periodic signals
- Time domain analysis of continuous time signalsSampling
- Sampling of continuous time signals usingMATLAB analyzing Signal Spectrum

Frequency domain analysis of continuous non-periodic signal

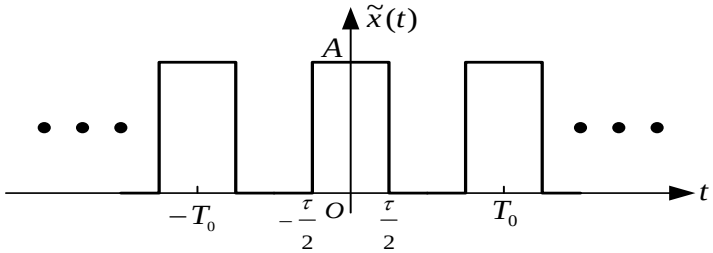
※Frequency domain representation of a continuous non-periodic signal

※Spectrum of a typical continuous non-periodic signal

※Continuous time Fourier transformProperties

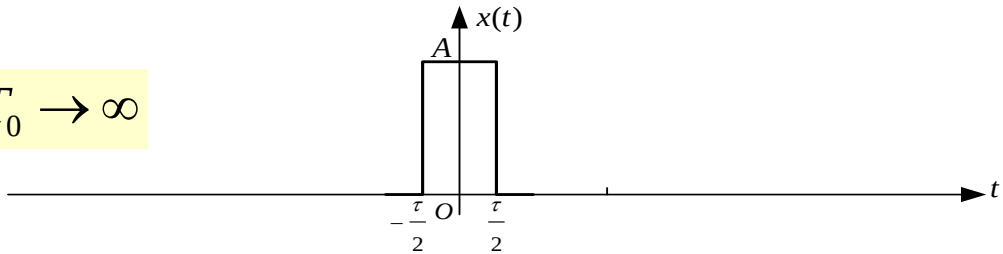
※Example of frequency domain analysis of a continuous non-periodic signal

The representation of continuous nonperiodic signals in frequency domain



periodic signal

$T_0 \to \infty$



Non-periodic signal

$$x(t) = \lim_{T_0 \to \infty} \tilde{x}(t)$$

Discuss Periodic T_0 increase on the discrete spectrum:

$$C_n = \frac{A\tau}{T_0} \mathrm{Sa}\left(\frac{n\omega_0 \tau}{2}\right)$$

$$\boxed{\lim_{T \to \infty} T_0 C_n = \lim_{T \to \infty} \frac{C_n}{f_0} = X(\mathrm{j}\omega)}$$

The representation of continuous nonperiodic signals in frequency domain

$$C_n = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

$$\lim_{T_0\to\infty} T_0 C_n = \lim_{T_0\to\infty}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t = \int_{-\infty}^{\infty}\lim_{T_0\to\infty}\tilde{x}(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

$$= \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = X(\mathrm{j}\omega)$$

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n\ \mathrm{e}^{\mathrm{j}n\omega_0 t} = \frac{1}{T_0}\sum_{n=-\infty}^{\infty} T_0 C_n\ \mathrm{e}^{\mathrm{j}n\omega_0 t} = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} T_0 C_n\ \mathrm{e}^{\mathrm{j}n\omega_0 t}\omega_0$$

$$x(t) = \lim_{T_0\to\infty}\tilde{x}(t) = \lim_{T_0\to\infty}\frac{1}{2\pi}\sum_{n=-\infty}^{\infty} T_0 C_n\ \mathrm{e}^{\mathrm{j}n\omega_0 t}\omega_0 = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\mathrm{j}\omega)\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega$$

The representation of continuous nonperiodic signals in frequency domain

※Fourier transform of a continuous non-periodic signal

$$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}X(\mathrm{j}\omega)\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega$$

$$X(\mathrm{j}\omega)=\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

- A signal can be decomposed into the linear combination of an infinite number of imaginary exponential signals $\mathrm{e}^{\mathrm{j}\omega t}$.
- The amplitude of the imaginary exponential signal $\mathrm{e}^{\mathrm{j}\omega t}$ is $\frac{|X(\mathrm{j}\omega)|}{2\pi}\mathrm{d}\omega$
- $X(\mathrm{j}w)$ is known as the spectrum function of the non-periodic signal.

$$X(\mathrm{j}\omega)=|X(\mathrm{j}\omega)|\mathrm{e}^{\mathrm{j}\varphi(\omega)}$$

Amplitude spectrum

Phase spectrum

The representation of continuous nonperiodic signals in frequency domain

The condition (Dirichlet condition) of a continuous non-periodic signal $x(t)$ exists a Fourier transform:

(1)A non-periodic signal is absolutely productible on its defining interval

$$\int_{-\infty}^{\infty} |x(t)| \mathrm{d}t < \infty$$

(2)The signal has only finitely many maxima and minima in any finite interval.

(3)A signal has only finitely many discontinuities in any finite interval and these discontinuities must be finite in value.

The representation of continuous nonperiodic signals in frequency domain

[Example**]**Calculate the Fourier transform of a non-periodic rectangular signal and plot the spectrum.

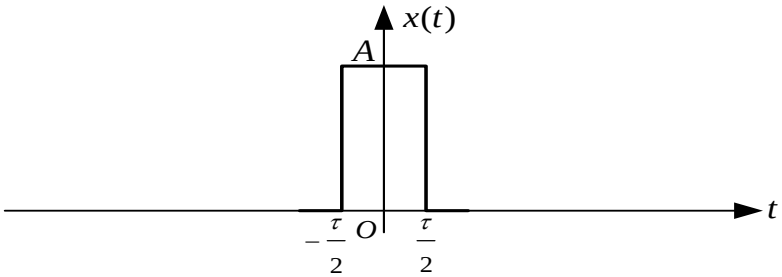
Solution:

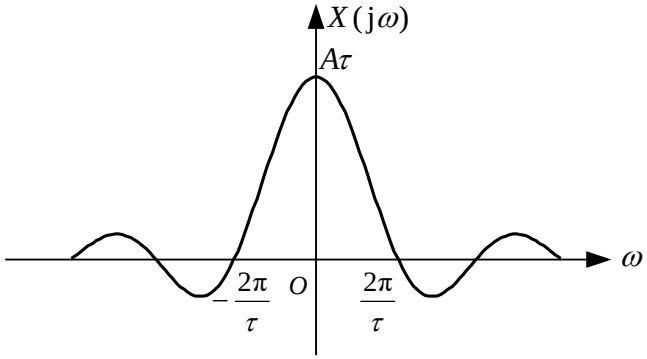
$$X(\mathrm{j}\omega)=\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$=\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}}A\cdot\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$=A\tau\cdot\mathrm{Sa}(\frac{\omega\tau}{2})$$

$$X(\mathrm{j}\omega)=A\tau\cdot\mathrm{Sa}(\frac{\omega\tau}{2})$$





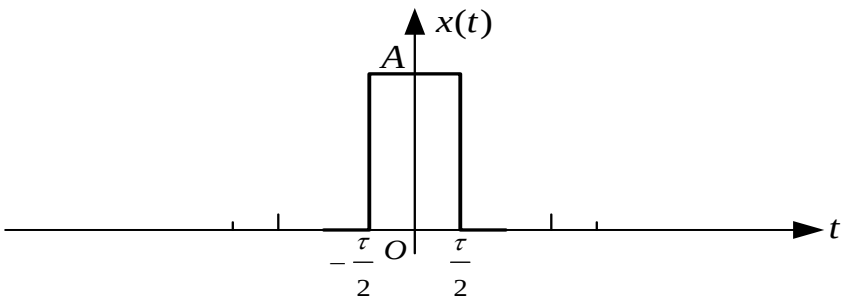
Spectrum of a non-periodic rectangular signal

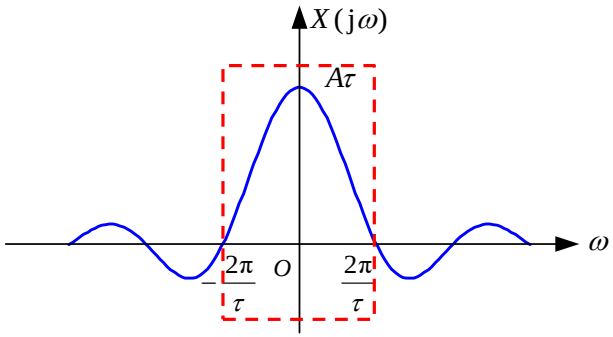
Frequency domain representation of a continuous non-periodic signal

※The Properties of continuous non-periodic signals

◆The spectrum of a non-periodic rectangular signal is continuous

◆If the spectrum of a signal in the time domain is finite, then in the frequency domain its spectrum continues to infinity





◆Effective bandwidth of the signal

The spectral components of a rectangular signal are mainly centered in (0~2π/τ).This frequency range is often referred to in engineering as the effective bandwidth ω_B of the signal in engineering is often called the effective bandwidth of the signal.

$$\omega_B=2\pi/\tau$$

Spectrum of a typical continuous non-periodic signal

Common continuous time signals

Unit Impulse Signals	DC Signals	Symbolic Function Signals
Unit Step Signal	Exponential Signal	Imaginary Exponential Signals
Cosine Signal	General Periodic Signal	Unit Impulse Series Signal

…

Spectrum of a typical continuous non-periodic signal

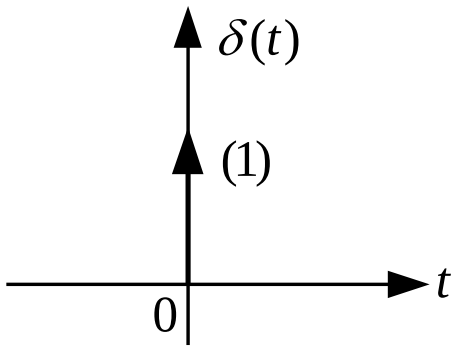
1.Unit impulse signal*d*(*t*)

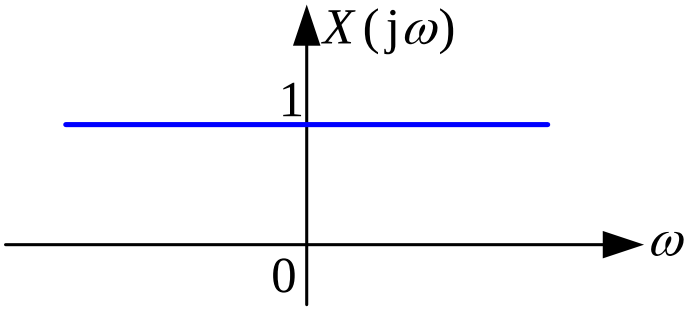
$$\mathscr{F}[\delta(t)] = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= \int_{-\infty}^{\infty} \delta(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= 1$$

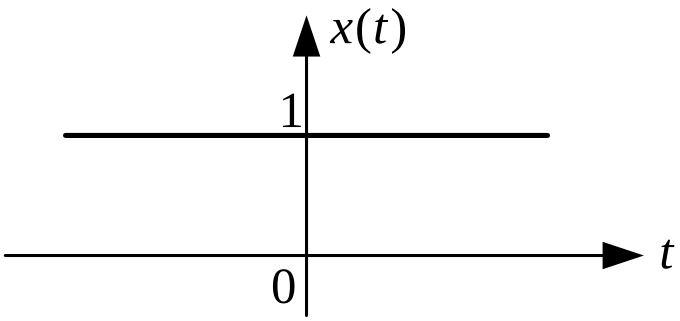
$$\mathscr{F}[\delta(t)] = 1$$





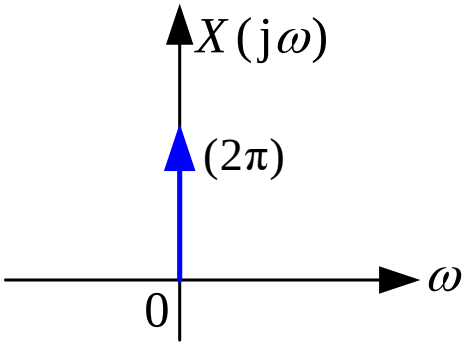
Spectrum of a typical continuous non-periodic signal

2.The DC signal $x(t) = 1$, -$-\infty<t<\infty$



$$\mathscr{F}[x(t)] = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= \int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = 2\pi\delta(\omega)$$



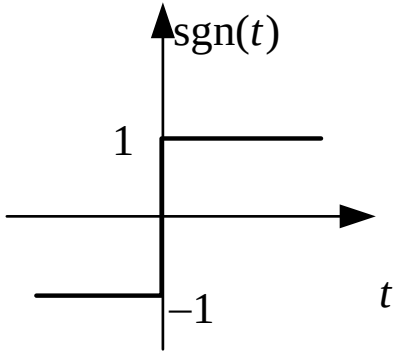
$$\delta(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 1\cdot \mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}\omega$$

$$\mathscr{F}[1] = 2\pi\delta(\omega)$$

Spectrum of a typical continuous non-periodic signal

3.Signal with sign function $\mathrm{sgn}(t)$

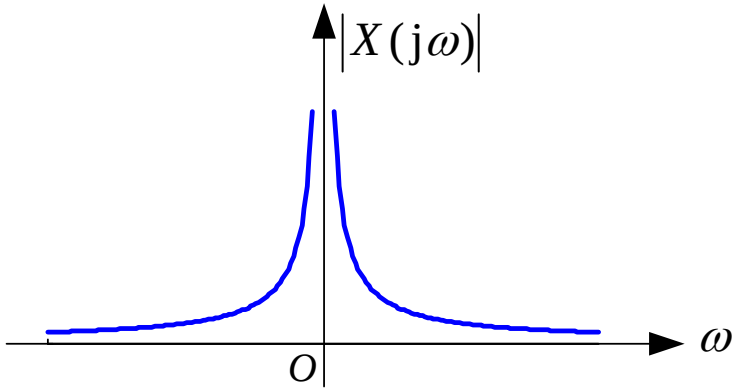
$$\mathrm{sgn}(t)=\begin{cases}-1 & t<0\\ 0 & t=0\\ 1 & t>0\end{cases}$$

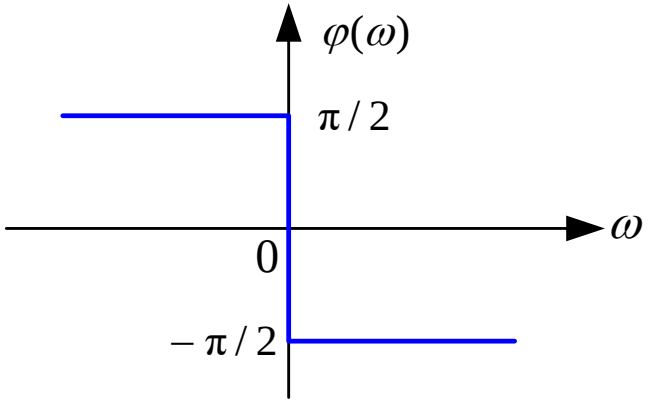


$$\mathscr{F}[\mathrm{sgn}(t)\mathrm{e}^{-\sigma|t|}]=\int_{-\infty}^{0}(-1)\mathrm{e}^{\sigma t}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t+\int_{0}^{\infty}\mathrm{e}^{-\sigma t}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t=\frac{-1}{\sigma-\mathrm{j}\omega}+\frac{1}{\sigma+\mathrm{j}\omega}$$

$$\mathrm{sgn}(t)=\lim_{\sigma\to 0}\mathrm{sgn}(t)\mathrm{e}^{-\sigma|t|}$$

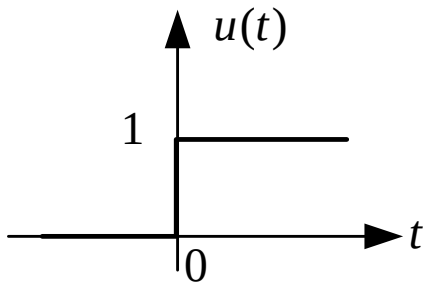
$$\mathscr{F}[\mathrm{sgn}(t)]=\frac{2}{\mathrm{j}\omega}$$





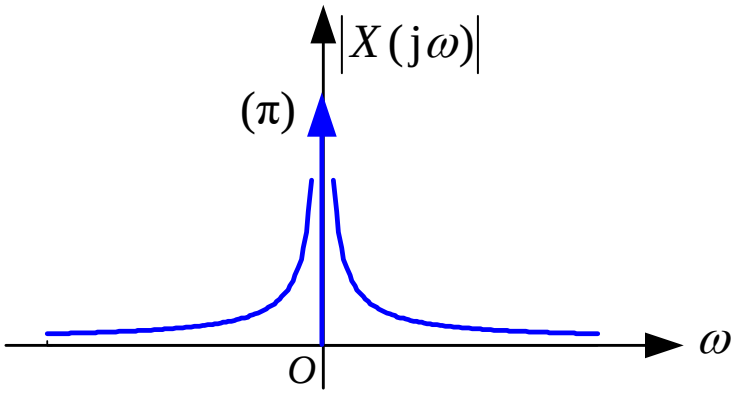
Spectrum of a typical continuous non-periodic signal

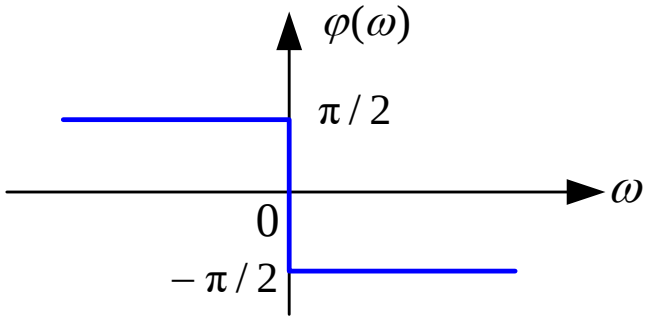
4.The unit step signal $u(t)$



$$u(t)=\frac{1}{2}+\frac{1}{2}\mathrm{sgn}(t)$$

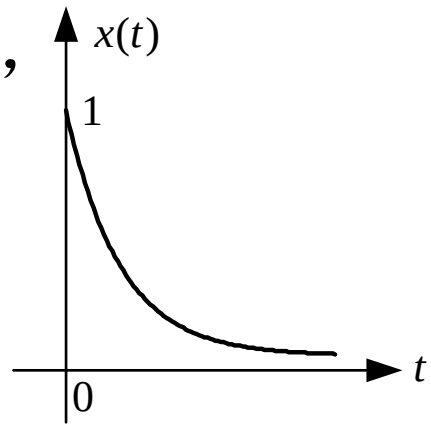
$$\mathscr{F}[u(t)]=\pi\delta(\omega)+\frac{1}{\mathrm{j}\omega}$$





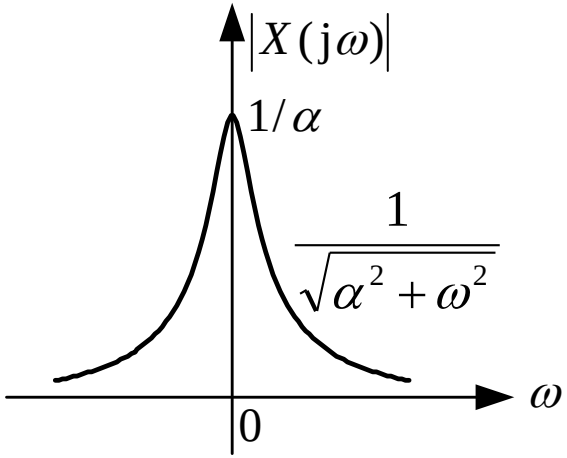
Spectrum of a typical continuous non-periodic signal

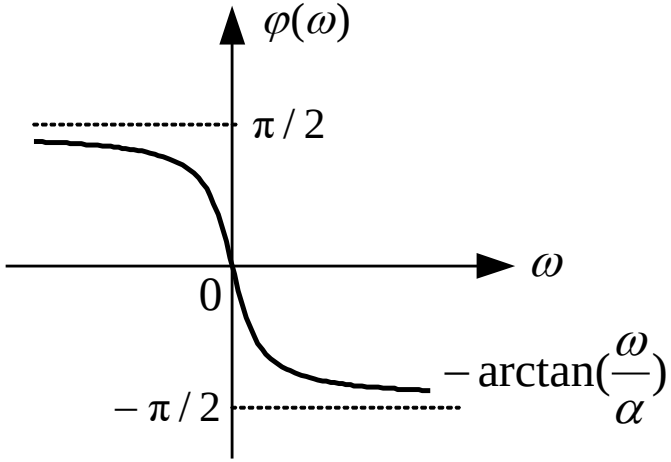
5.Unilateral exponential signal $\mathrm{e}^{-\alpha t}u(t)$, $\alpha > 0$,



$$X(\mathrm{j}\omega) = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t = \int_{0}^{\infty} \mathrm{e}^{-\alpha t}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= \frac{\mathrm{e}^{-(\alpha+\mathrm{j}\omega)t}}{-(\alpha+\mathrm{j}\omega)}\Bigg|_{0}^{\infty} = \frac{1}{\alpha+\mathrm{j}\omega}$$



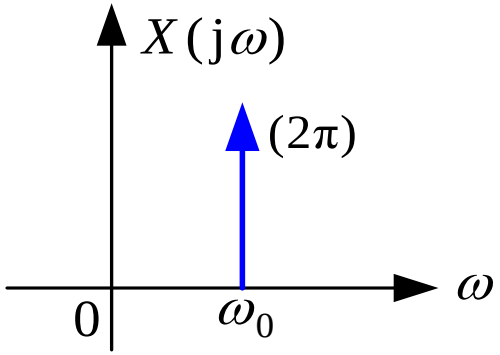


Spectrum of a typical continuous non-periodic signal

6.Imaginary Exponential Signal $\mathrm{e}^{\mathrm{j}\omega_0 t}(-\infty < t < \infty)$

$$\mathscr{F}[\mathrm{e}^{\mathrm{j}\omega_0 t}] = \int_{-\infty}^{\infty} \mathrm{e}^{-\mathrm{j}(\omega-\omega_0)t}\mathrm{d}t$$

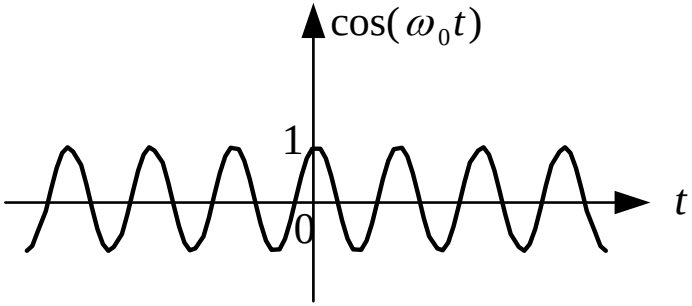
$$= 2\pi\delta(\omega-\omega_0)$$



$$\mathscr{F}[\mathrm{e}^{\mathrm{j}\omega_0 t}] = 2\pi\delta(\omega-\omega_0)$$

Spectrum of a typical continuous non-periodic signal

7.Cosine Signal $\cos(\omega_0 t)$

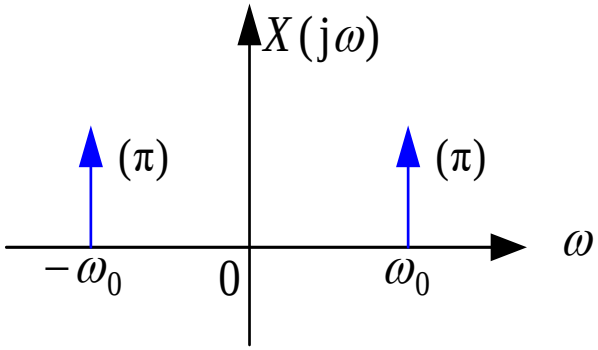


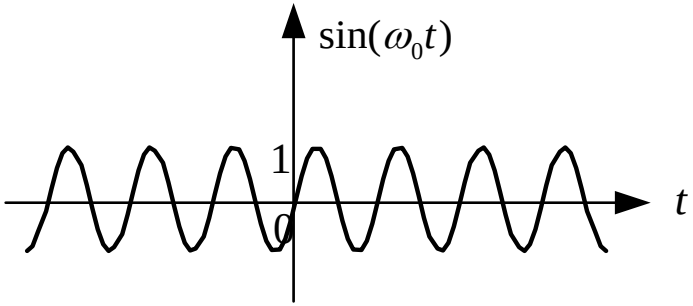
$$\cos(\omega_0 t) = \frac{1}{2}(\mathrm{e}^{\mathrm{j}\omega_0 t} + \mathrm{e}^{-\mathrm{j}\omega_0 t})$$

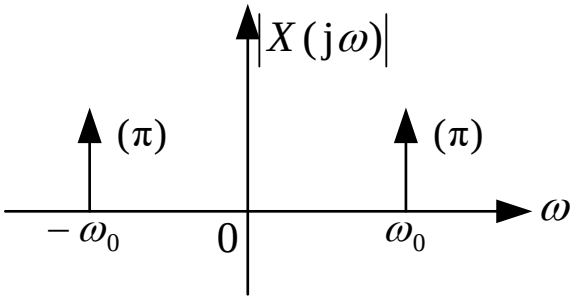
$$\mathscr{F}[\mathrm{e}^{\mathrm{j}\omega_0 t}] = 2\pi\delta(\omega - \omega_0)$$

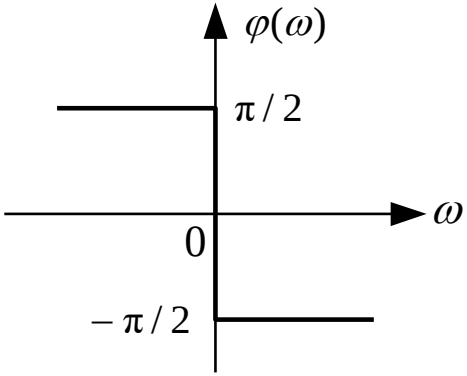
$$\mathscr{F}[\cos\omega_0 t] = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$\mathscr{F}[\sin\omega_0 t] = \mathrm{j}\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$









Spectrum of a typical continuous non-periodic signal

8.General Periodic Signal

$$\tilde{x}(t)=\sum_{n=-\infty}^{+\infty}C_n \mathrm{e}^{\mathrm{j}n\omega_0 t} \qquad (\omega_0=\frac{2\pi}{T_0})$$

$$\mathscr{F}[\tilde{x}(t)]=X(\mathrm{j}\omega)=\mathscr{F}[\sum_{n=-\infty}^{+\infty}C_n \mathrm{e}^{\mathrm{j}n\omega_0 t}]$$

$$=\sum_{n=-\infty}^{+\infty}C_n\cdot\mathscr{F}[\mathrm{e}^{\mathrm{j}n\omega_0 t}]$$

$$\mathscr{F}[\tilde{x}(t)]=2\pi\sum_{n=-\infty}^{+\infty}C_n\delta(\omega-n\omega_0)$$

Spectrum of a typical continuous non-periodic signal

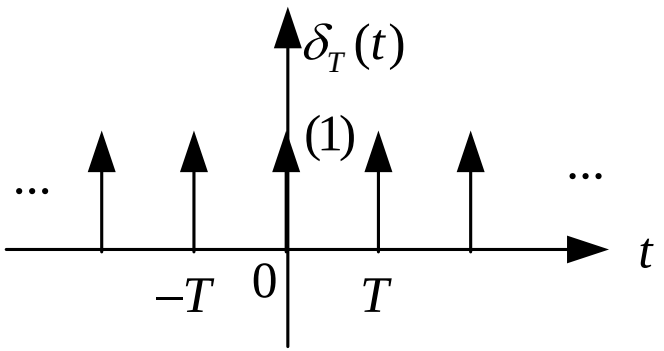
9.Unit impulse series $\delta_T(t)$

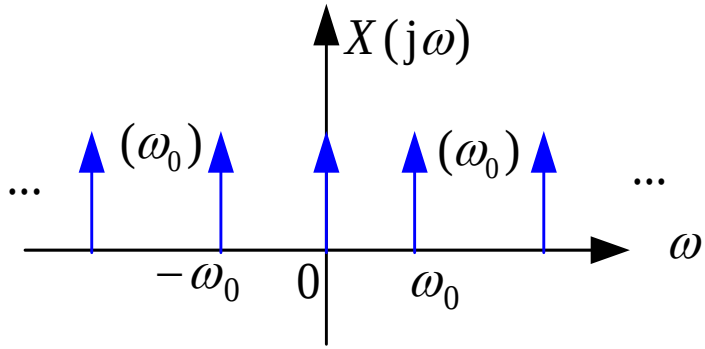
$$\delta_T(t)=\sum_{n=-\infty}^{+\infty}\delta(t-nT)=\sum_{n=-\infty}^{+\infty}C_n\mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$C_n=\frac{1}{T}\int_{-T/2}^{T/2}\delta(t)\mathrm{e}^{-\mathrm{j}n\omega_0 t}\mathrm{d}t$$

$$=\frac{1}{T}\mathrm{e}^{-\mathrm{j}n\omega_0 t}\Big|_{t=0}=\frac{1}{T}$$

$$\mathscr{F}[\delta_T(t)]=2\pi\sum_{n=-\infty}^{+\infty}\frac{1}{T}\delta(\omega-n\omega_0)$$





$$\mathscr{F}[\delta_T(t)]=\omega_0\sum_{n=-\infty}^{+\infty}\delta(\omega-n\omega_0)$$

Properties of the continuous-time Fourier transform

- Linear Characterization
- Conjugate Symmetry Properties
- Reciprocal Symmetry Properties
- Expansion Property
- Time-shift Property
- Frequency Shift Property
- Convolution Property
- Product Property
- Time Domain Differential Properties
- Integral Property
- Frequency Differential Properties
- Energy Conservation Theorem

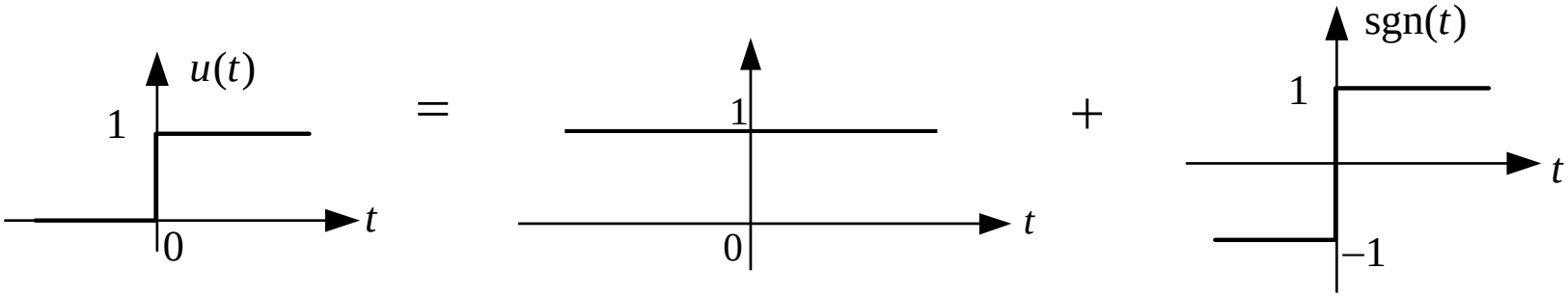
Properties of the continuous-time Fourier transform

1.Linear Properties

If $x_1(t) \xleftrightarrow{\mathscr{F}} X_1(\mathrm{j}\omega)$; $x_2(t) \xleftrightarrow{\mathscr{F}} X_2(\mathrm{j}\omega)$

then $ax_1(t)+bx_2(t) \xleftrightarrow{\mathscr{F}} aX_1(\mathrm{j}\omega)+bX_2(\mathrm{j}\omega)$

Where a and b are constants



$$u(t)=\frac{1}{2}+\frac{1}{2}\mathrm{sgn}(t) \qquad \mathscr{F}[1]=2\pi\delta(\omega) \qquad \mathscr{F}[\mathrm{sgn}(t)]=\frac{2}{\mathrm{j}\omega}$$

$$\mathscr{F}[u(t)]=\pi\delta(\omega)+\frac{1}{\mathrm{j}\omega}$$

Properties of continuous time Fourier transform

2.Conjugate Symmetry Property

If $$x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$$

then $$x^*(t) \xleftrightarrow{\mathscr{F}} X^*(-\mathrm{j}\omega) \qquad x^*(-t) \xleftrightarrow{\mathscr{F}} X^*(\mathrm{j}\omega)$$

$X(\mathrm{j}\omega)$ is a complex number and can be expressed as

$$X(\mathrm{j}\omega) = |X(\mathrm{j}\omega)|\mathrm{e}^{\mathrm{j}\varphi(\omega)} = X_{\mathrm{R}}(\mathrm{j}\omega) + \mathrm{j}X_{\mathrm{I}}(\mathrm{j}\omega)$$

(1)When $\boldsymbol{x(t)}$ is a real signal $\quad X(\mathrm{j}\omega) = X^*(-\mathrm{j}\omega)$

$$|X(\mathrm{j}\omega)|\mathrm{e}^{\mathrm{j}\varphi(\omega)} = |X(-\mathrm{j}\omega)|\mathrm{e}^{-\mathrm{j}\varphi(-\omega)}$$

$$|X(\mathrm{j}\omega)| = |X(-\mathrm{j}\omega)| \qquad \varphi(\omega) = -\varphi(-\omega)$$

$$X_{\mathrm{R}}(\mathrm{j}\omega) = X_{\mathrm{R}}(-\mathrm{j}\omega), \quad X_{\mathrm{I}}(-\mathrm{j}\omega) = -X_{\mathrm{I}}(-\mathrm{j}\omega)$$

Properties of the continuous time Fourier transform

2.Conjugate Symmetry Property

(2)When $\boldsymbol{x(t)}$ is a real even signal $\quad x(t) = x^*(-t)$

$X(\mathrm{j}\omega) = X^*(\mathrm{j}\omega)$ $\quad X(\mathrm{j}\omega)$ is real even function

(3)When $\boldsymbol{x(t)}$ is a real odd signal $\quad x(t) = -x^*(-t)$

$X(\mathrm{j}\omega) = -X^*(\mathrm{j}\omega)$ $\quad X(\mathrm{j}\omega)$ is a purely imaginary odd function

$$x(t) = \boxed{x_{\mathrm{e}}(t)} + \boxed{x_{\mathrm{o}}(t)}$$

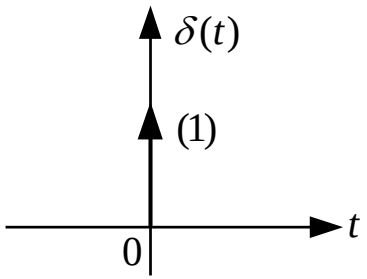
$$\Downarrow \qquad\qquad \Downarrow$$

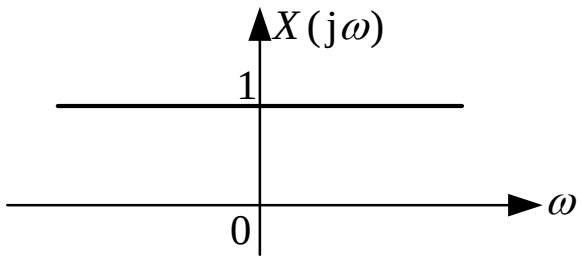
$$X_{\mathrm{R}}(\mathrm{j}\omega) \qquad X_{\mathrm{I}}(\mathrm{j}\omega)$$

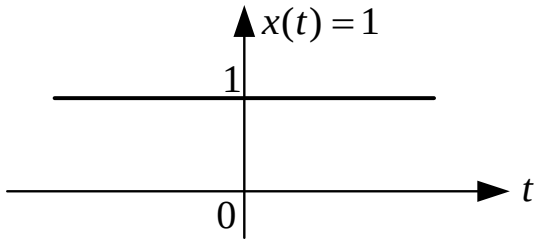
Properties of the continuous-time Fourier transform

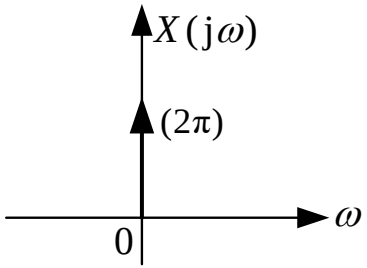
3.Reciprocal Symmetry Property

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ Then $X(\mathrm{j}t) \xleftrightarrow{\mathscr{F}} 2\pi \cdot x(-\omega)$





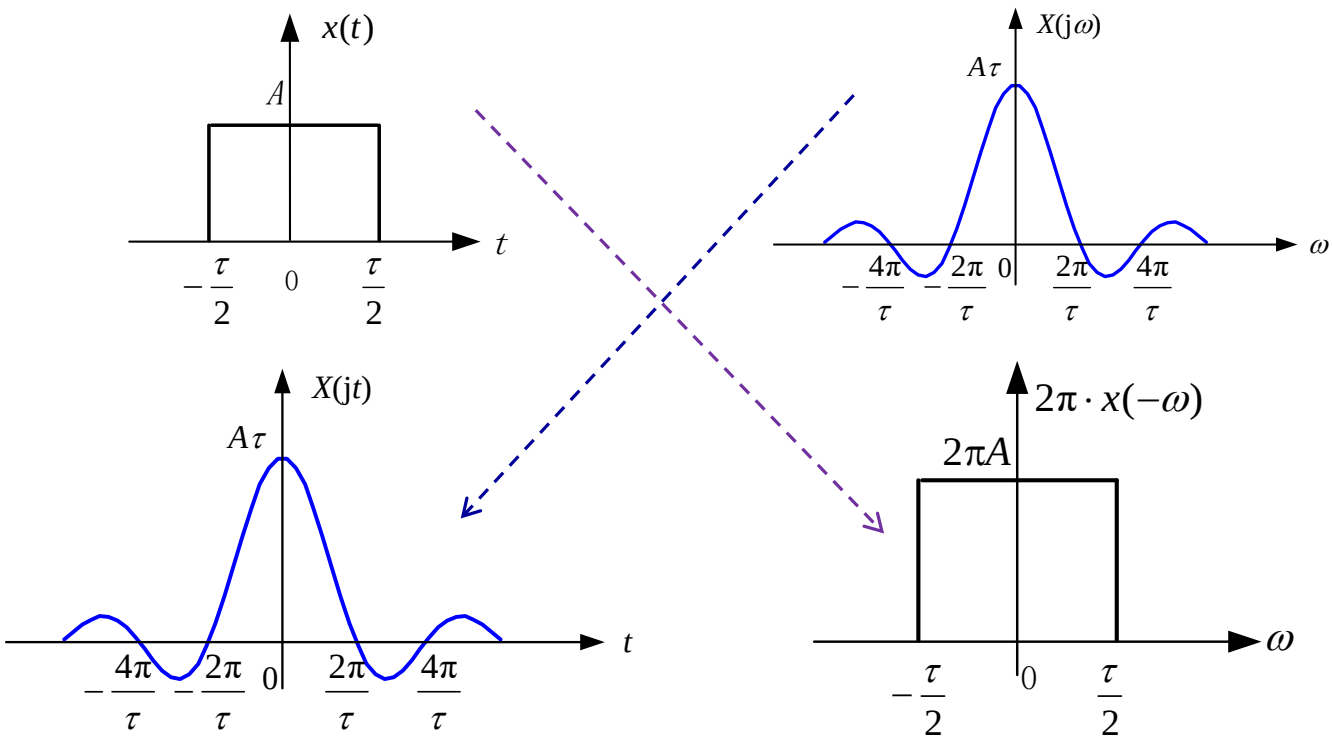




Properties of the continuous time Fourier transform

3.Reciprocal Symmetry Property

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ then $X(\mathrm{j}t) \xleftrightarrow{\mathscr{F}} 2\pi \cdot x(-\omega)$



Properties of continuous time Fourier transform

4.Expansion and Contraction Property

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ then $x(at) \xleftrightarrow{\mathscr{F}} \frac{1}{|a|} X(\mathrm{j}\frac{\omega}{a})$

Prove that: $\mathscr{F}[x(at)] = \int_{-\infty}^{\infty} x(at)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$

Let $\lambda = at$ then $\mathrm{d}\lambda = a\mathrm{d}t$ which can be obtained by substituting into the above equation

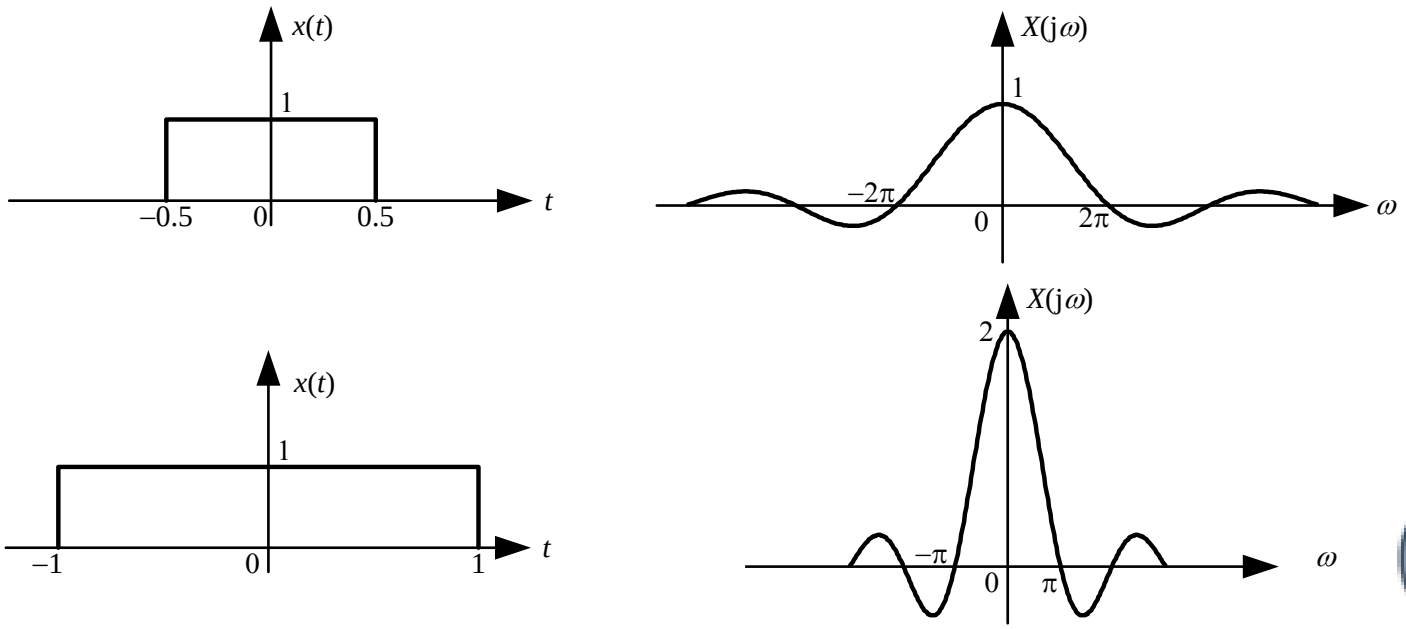
$$\mathscr{F}[x(at)] = \frac{1}{|a|}\int_{-\infty}^{\infty} x(\lambda)\mathrm{e}^{-\mathrm{j}\frac{\omega}{a}\lambda}\mathrm{d}\lambda = \frac{1}{|a|} X(\mathrm{j}\frac{\omega}{a})$$

If the time domain is compressed, the frequency domain is broadened; if the time domain is broadened, the frequency domain is compressed.

Properties of continuous time Fourier transform

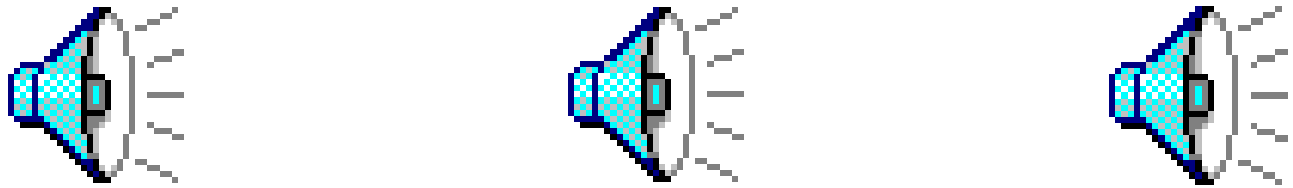
4.Expansion and Contraction Property

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ then $x(at) \xleftrightarrow{\mathscr{F}} \frac{1}{|a|} X(\mathrm{j}\frac{\omega}{a})$

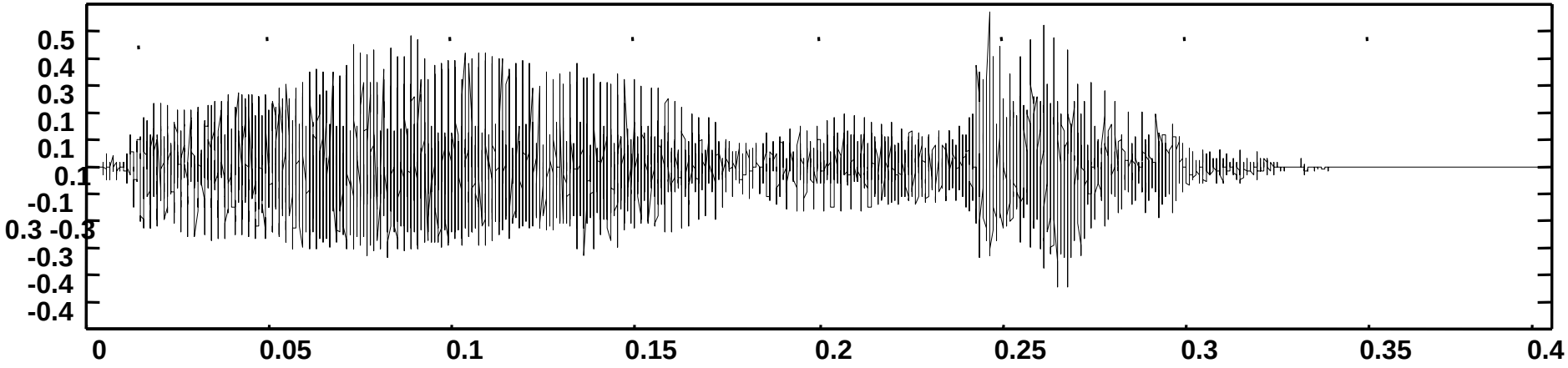


Properties of continuous time Fourier transform

Scale transformation changes in the speech signal after



$x(t)$ $x(2t)$ $x(t/2)$



A voice signal(“That's right.”).
Sampling frequency= 22050Hz

Properties of the continuous time Fourier transform

5.Property of Time Shift

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ then $x(t-t_0) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)\mathrm{e}^{-\mathrm{j}\omega t_0}$

$$X(\mathrm{j}\omega)\cdot \mathrm{e}^{-\mathrm{j}\omega t_0} = |X(\mathrm{j}\omega)|\,\mathrm{e}^{\mathrm{j}[\varphi(\omega)-\omega t_0]}$$ t_0 is any real number

The time shift of the signal in the time domain corresponds to the phase shift of the spectral function in the frequency domain.

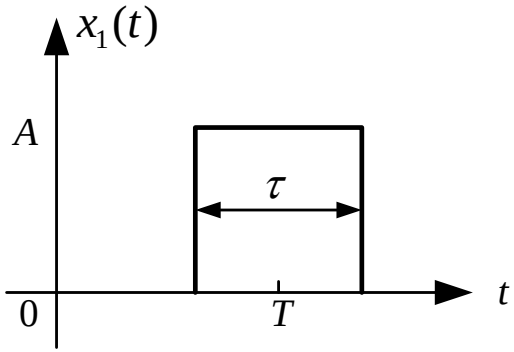
Proof: $$\mathscr{F}[x(t-t_0)] = \int_{-\infty}^{\infty} x(t-t_0)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

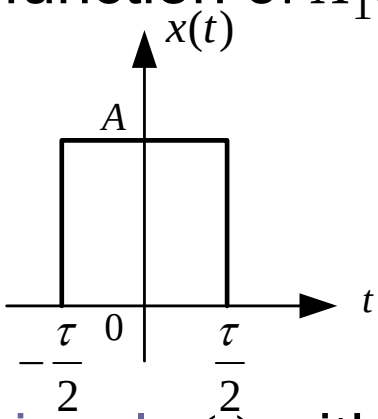
Let $x = t - t_0$ so $\mathrm{d}x = \mathrm{d}t$, which is the same as the one in the above equation. Substituting into the above equation yields.

$$\mathscr{F}[x(t-t_0)] = \int_{-\infty}^{\infty} x(\lambda)\mathrm{e}^{-\mathrm{j}\omega(t_0+\lambda)}\mathrm{d}\lambda = X(\mathrm{j}\omega)\cdot \mathrm{e}^{-\mathrm{j}\omega t_0}$$

Properties of continuous time Fourier transform

Ex. Find the time-delayed rectangular pulse signal shown in the figure $x_1(t)$ and the spectral function of $X_1(\mathrm{j}\omega)$.





Solution: The rectangular pulse signal $x(t)$ without delay and with width τ as shown in the figure,

The corresponding spectral function is $X(\mathrm{j}\omega) = A\tau \cdot \mathrm{Sa}(\frac{\omega\tau}{2})$

Since $x_1(t) = x(t-T)$

Therefore, by the delay characteristic it can be obtained that

$$X_1(\mathrm{j}\omega) = X(\mathrm{j}\omega)\mathrm{e}^{-\mathrm{j}\omega T} = A\tau \cdot \mathrm{Sa}(\frac{\omega\tau}{2})\mathrm{e}^{-\mathrm{j}\omega T}$$

6.Frequency Shift Property

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ then $x(t)\cdot \mathrm{e}^{\mathrm{j}\omega_0 t} \xleftrightarrow{\mathscr{F}} X[\mathrm{j}(\omega-\omega_0)]$

The phase shift of the signal in the time domain corresponds to the frequency shift of the spectral function in the frequency domain.

Proof: From the definition of Fourier transform

$$\mathscr{F}[x(t)\cdot \mathrm{e}^{\mathrm{j}\omega_0 t}] = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{\mathrm{j}\omega_0 t}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$= \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}(\omega-\omega_0)t}\mathrm{d}t$$

$$= X[\mathrm{j}(\omega-\omega_0)]$$

Properties of the continuous time Fourier transform

6.Frequency Shift Property

If $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ then $x(t)\cdot \mathrm{e}^{\mathrm{j}\omega_0 t} \xleftrightarrow{\mathscr{F}} X[\mathrm{j}(\omega-\omega_0)]$

$$\mathscr{F}[x(t)\cos\omega_0 t] = \frac{1}{2}\mathscr{F}[x(t)\mathrm{e}^{\mathrm{j}\omega_0 t}] + \frac{1}{2}\mathscr{F}[x(t)\mathrm{e}^{-\mathrm{j}\omega_0 t}]$$

$$= \frac{1}{2}X[\mathrm{j}(\omega-\omega_0)] + \frac{1}{2}X[\mathrm{j}(\omega+\omega_0)]$$

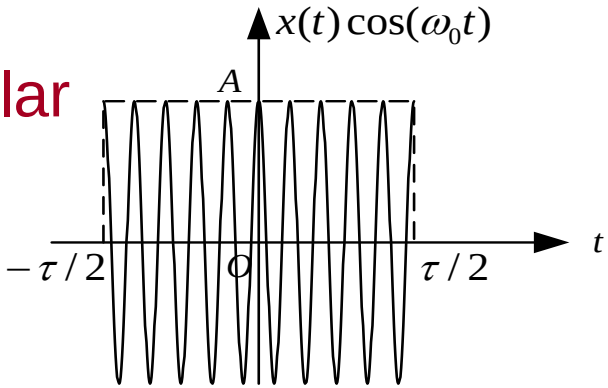
The signal $x(t)$ and the cosine signal $\cos\omega_0 t$ multiplied by the cosine signal, the spectrum is shifted to the left and right of the original signal spectrum ω_0, theThe amplitude is halved.

Similarly $\mathscr{F}[x(t)\sin\omega_0 t] = \frac{1}{2\mathrm{j}}\mathscr{F}[x(t)\mathrm{e}^{\mathrm{j}\omega_0 t}] - \frac{1}{2\mathrm{j}}\mathscr{F}[x(t)\mathrm{e}^{-\mathrm{j}\omega_0 t}]$

$$= -\frac{\mathrm{j}}{2}X[\mathrm{j}(\omega-\omega_0)] + \frac{\mathrm{j}}{2}X[\mathrm{j}(\omega+\omega_0)]$$

Properties of the continuous-time Fourier transform

[Ex.] Try to find the spectral function of the signal after multiplying the rectangular signal $x(t)$ and the cosine signal $\cos(\omega_0 t)$.



Solution:The spectral function of a rectangular signal of width τ is

$$X(\mathrm{j}\omega) = A\tau \cdot \mathrm{Sa}(\frac{\omega\tau}{2})$$

Applying the frequency shift property can be obtained

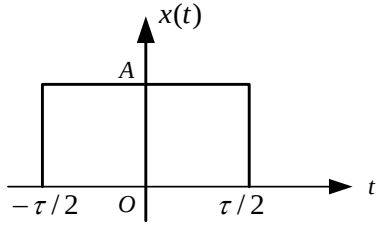
$$\mathscr{F}[x(t)\cos(\omega_0 t)] = \frac{1}{2}X[\mathrm{j}(\omega-\omega_0)] + \frac{1}{2}X[\mathrm{j}(\omega+\omega_0)]$$

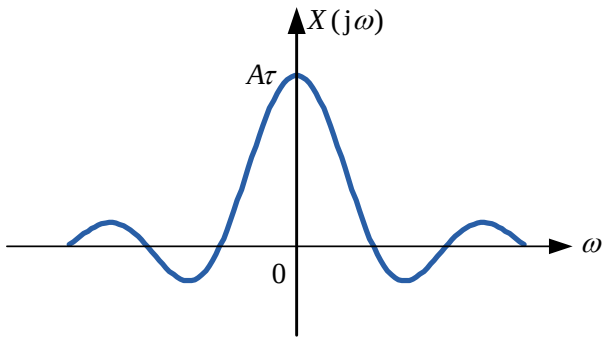
$$= \frac{A\tau}{2}\{\mathrm{Sa}\frac{(\omega-\omega_0)\tau}{2} + \mathrm{Sa}\frac{(\omega+\omega_0)\tau}{2}\}$$

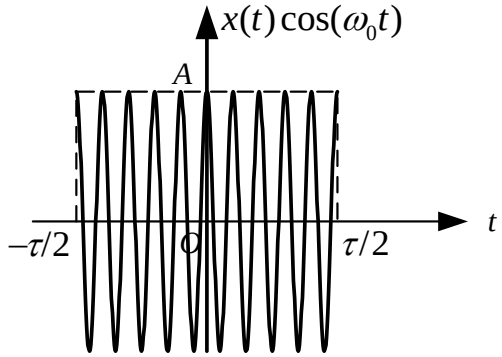
Properties of the continuous time Fourier transform

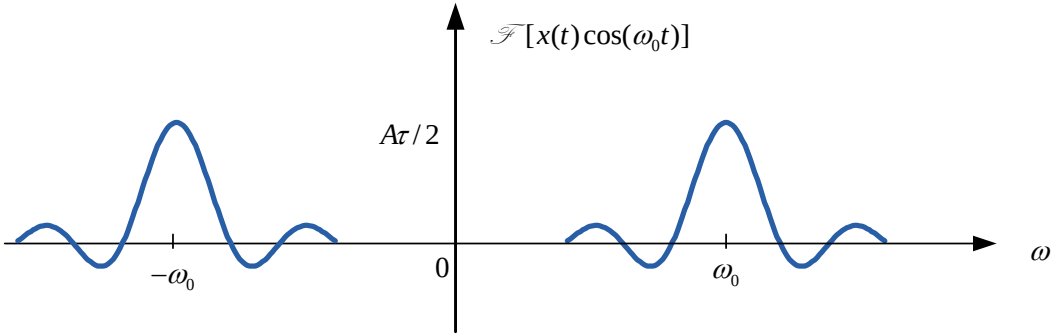
[Ex.] Try to find the spectral function of the signal after multiplying the rectangular signal $x(t)$ and the cosine signal $\cos(\omega_0 t)$.

Solution:









Properties of the continuous time Fourier transform

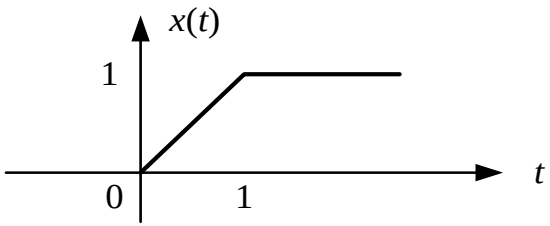
7.Time domainIntegral Property

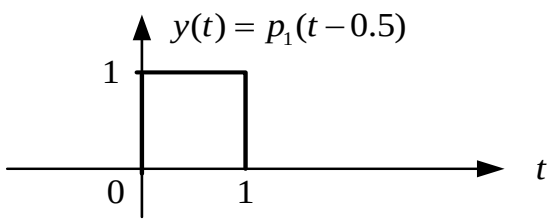
$$若 x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$$

$$则 \int_{-\infty}^{t} x(\tau)\mathrm{d}\tau \xleftrightarrow{\mathscr{F}} \frac{1}{\mathrm{j}\omega} X(\mathrm{j}\omega) + \pi X(0)\delta(\omega)$$

Properties of the continuous time Fourier transform

[Ex.]Try to utilize the Integral property to find the spectrum of signal shown in the graph $x(t)$.





Solution: $$x(t)=\int_{-\infty}^{t} y(t)\mathrm{d}t=\int_{-\infty}^{t} p_1(t-0.5)\mathrm{d}t$$

$$p_1(t-0.5)\xleftrightarrow{\mathscr{F}}\mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega}$$

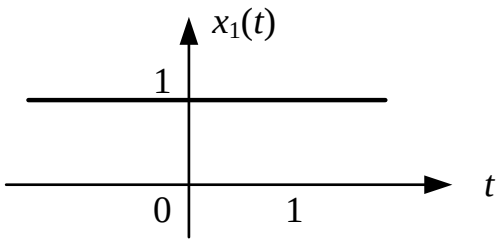
Utilize the time-domain integral characteristic, we can obtain that

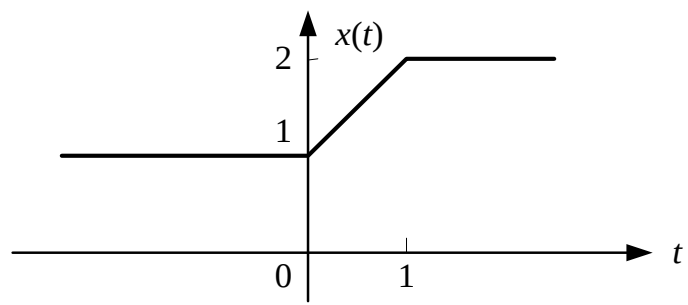
$$X(\mathrm{j}\omega)=\frac{1}{\mathrm{j}\omega}\mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega}+\pi\delta(\omega)$$

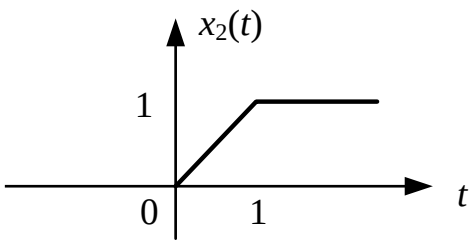
Properties of the continuous time Fourier transform

[Ex.]Try to use the integral property to find the spectrum of the graphical signal $x(t)$.

Solution: Denote $x(t)$ as $x_1(t)+x_2(t)$







$$x(t)=1+\int_{-\infty}^{t} p(t-0.5)\mathrm{d}t$$

$$X(\mathrm{j}\omega)=\frac{1}{\mathrm{j}\omega}\mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega}+3\pi\delta(\omega)$$

8.Property of time-domain differentiation

If

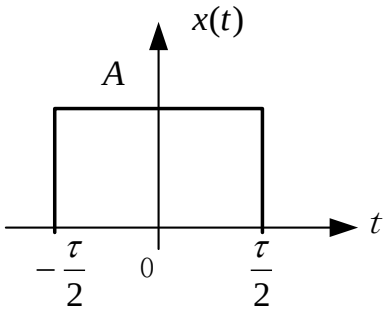
$$x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$$

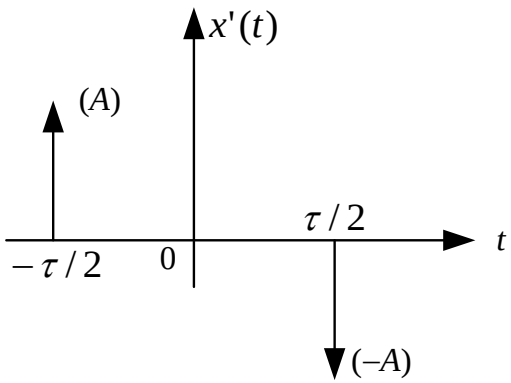
then

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} \xleftrightarrow{\mathscr{F}} (\mathrm{j}\omega)\cdot X(\mathrm{j}\omega) \qquad \frac{\mathrm{d}^n x(t)}{\mathrm{d}t^n} \xleftrightarrow{\mathscr{F}} (\mathrm{j}\omega)^n \cdot X(\mathrm{j}\omega)$$

Properties of the continuous time Fourier transform

[Ex.]Try to utilize the Differential Property to find the spectrum of a rectangular signal.





Solution: $x'(t) = A\delta(t+\frac{\tau}{2}) - A\delta(t-\frac{\tau}{2})$

$$\mathscr{F}[x'(t)] = A\mathrm{e}^{\mathrm{j}\omega\frac{\tau}{2}} - A\mathrm{e}^{-\mathrm{j}\omega\frac{\tau}{2}} = A\cdot 2\mathrm{j}\sin(\omega\frac{\tau}{2})$$

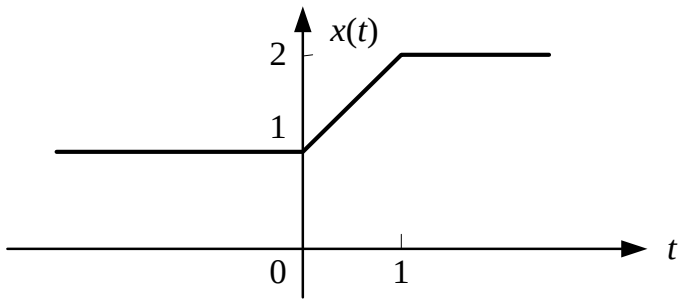
Using the time-domain differential characteristic, we have

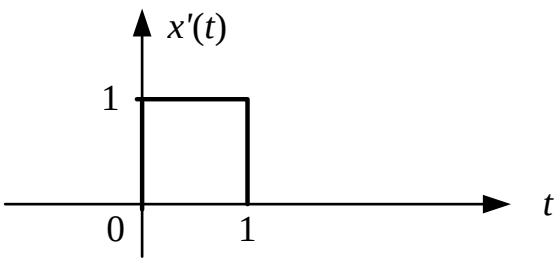
$$\mathscr{F}[x'(t)] = (\mathrm{j}\omega)X(\mathrm{j}\omega) = A\cdot 2\mathrm{j}\sin(\omega\frac{\tau}{2})$$

Thus there is $$X(\mathrm{j}\omega) = \frac{2A}{\omega}\sin(\omega\frac{\tau}{2}) = A\tau\cdot\mathrm{Sa}(\frac{\omega\tau}{2})$$

Properties of the continuous-time Fourier transform

[Ex]Try to utilize the Differential Property to find the spectrum of the graphical signal $x(t)$.





Solution: $$x'(t) = p_1(t-0.5) \xleftrightarrow{\mathscr{F}} \mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega}$$

Using the time-domain differential characteristic

$$X(\mathrm{j}\omega) = \frac{1}{\mathrm{j}\omega}\mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega} \neq \frac{1}{\mathrm{j}\omega}\mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega} + 3\pi\delta(\omega)$$

The time-domain differentiation of the signal causes the DC component of the signal to be lost.

Properties of the continuous time Fourier transform

8. Modified time domain differential property

$$x'(t) = x_1(t)$$

若 $x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$ $\quad x_1(t) \xleftrightarrow{\mathscr{F}} X_1(\mathrm{j}\omega)$

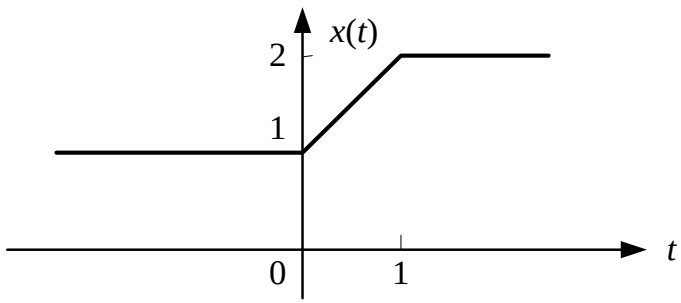
then

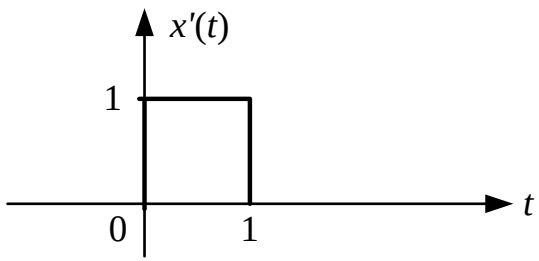
$$X(\mathrm{j}\omega) = \pi[x(+\infty) + x(-\infty)]\delta(\omega) + \frac{X_1(\mathrm{j}\omega)}{\mathrm{j}\omega}$$

Properties of the continuous time Fourier transform

[Ex.] Try to utilize the modified differential characteristic to find the spectral function of a rectangular pulse signal

Solution:





$$x'(t) = p_1(t-0.5) = x_1(t) \xleftrightarrow{\mathscr{F}} X_1(\mathrm{j}\omega) = \mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega}$$

Using the modified differential characteristic:

$$X(\mathrm{j}\omega) = \pi[x(+\infty) + x(-\infty)]\delta(\omega) + \frac{X_1(\mathrm{j}\omega)}{\mathrm{j}\omega}$$

$$= 3\pi\delta(\omega) + \frac{1}{\mathrm{j}\omega}\mathrm{Sa}(0.5\omega)\mathrm{e}^{-\mathrm{j}0.5\omega}$$

Properties of continuous time Fourier transform

9.Differential property in frequency domain

If $$x(t) \xleftrightarrow{\mathscr{F}} X(\mathrm{j}\omega)$$

then $$t \cdot x(t) \xleftrightarrow{\mathscr{F}} \mathrm{j}\frac{\mathrm{d}X(\mathrm{j}\omega)}{\mathrm{d}\omega} \qquad t^n x(t) \xleftrightarrow{\mathscr{F}} \mathrm{j}^n \frac{\mathrm{d}X^n(\mathrm{j}\omega)}{\mathrm{d}\omega^n}$$

Proof: $$X(\mathrm{j}\omega) = \int_{-\infty}^{\infty} x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

$$\frac{\mathrm{d}X(\mathrm{j}\omega)}{\mathrm{d}\omega} = \int_{-\infty}^{\infty} x(t)\frac{\mathrm{d}}{\mathrm{d}\omega}\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}\omega = \int_{-\infty}^{\infty}[(-\mathrm{j}t)x(t)]\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

We multiply both sides of this equation by j

$$\mathrm{j}\frac{\mathrm{d}X(\mathrm{j}\omega)}{\mathrm{d}\omega} = \int_{-\infty}^{\infty}[tx(t)]\cdot\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

Properties of the continuous time Fourier transform

[Ex.] Try to find the frequency spectrum of the unit ramp signal $\boldsymbol{tu(t)}$

Solution: It is known that the Fourier transform of the unit step signal is

$$\mathscr{F}[u(t)] = \pi\delta(\omega) + \frac{1}{\mathrm{j}\omega}$$

Therefore, by using the frequency domain differential characteristics, we can obtain:

$$\mathscr{F}[tu(t)] = \mathrm{j}\frac{\mathrm{d}}{\mathrm{d}\omega}[\pi\delta(\omega) + \frac{1}{\mathrm{j}\omega}] = \mathrm{j}\pi\delta'(\omega) - \frac{1}{\omega^2}$$

Properties of continuous time Fourier transform

10.Time domain convolution property

If $x_1(t) \xleftrightarrow{\mathscr{F}} X_1(\mathrm{j}\omega)$ $\quad x_2(t) \xleftrightarrow{\mathscr{F}} X_2(\mathrm{j}\omega)$

then $x_1(t) * x_2(t) \xleftrightarrow{\mathscr{F}} X_1(\mathrm{j}\omega) \cdot X_2(\mathrm{j}\omega)$

Proof:

$$\mathscr{F}[x_1(t) * x_2(t)] = \int_{-\infty}^{\infty} [\int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau) \mathrm{d}\tau] \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t$$

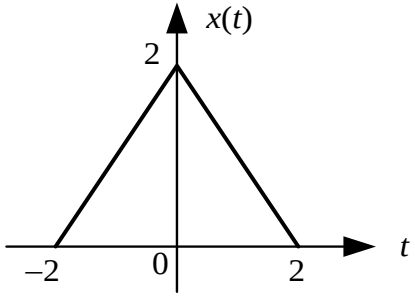
$$= \int_{-\infty}^{\infty} x_1(\tau) [\int_{-\infty}^{\infty} x_2(t-\tau) \mathrm{e}^{-\mathrm{j}\omega t} \mathrm{d}t] \mathrm{d}\tau$$

$$= \int_{-\infty}^{\infty} x_1(\tau) X_2(\mathrm{j}\omega) \mathrm{e}^{-\mathrm{j}\omega\tau} \mathrm{d}\tau$$

$$= X_1(\mathrm{j}\omega) \cdot X_2(\mathrm{j}\omega)$$

Properties of continuous time Fourier transform

[Ex.]Find the spectrum of the signal shown in the figure.

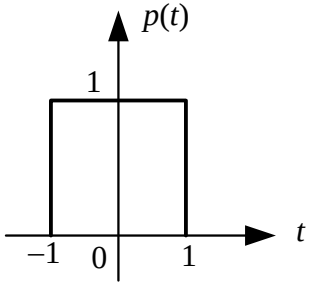


Solution:

$$x(t) = p_2(t) * p_2(t) \qquad p_2(t) \xleftrightarrow{\mathscr{F}} 2\mathrm{Sa}(\omega)$$

According to the time domain convolution property

$$X(\mathrm{j}\omega) = 4\mathrm{Sa}^2(\omega)$$



Properties of the continuous time Fourier transform

[Ex.]Find the spectrum of the signal

$$y(t)=\int_{-2}^{t} \mathrm{e}^{-2\tau}\cdot \mathrm{e}^{-5(t-\tau)}\mathrm{d}\tau,\ (t>-2).$$

Solution:

$$y(t)=\int_{-2}^{t} \mathrm{e}^{-2\tau}\cdot \mathrm{e}^{-5(t-\tau)}\mathrm{d}\tau=\mathrm{e}^{-2t}u(t+2)*\mathrm{e}^{-5t}u(t)$$

Using the convolution property of the Fourier transform we can obtain

$$Y(\mathrm{j}\omega)=\mathscr{F}[\mathrm{e}^{-2t}u(t+2)]\mathscr{F}[\mathrm{e}^{-5t}u(t)]$$

$$=\frac{\mathrm{e}^{4}\mathrm{e}^{\mathrm{j}2\omega}}{\mathrm{j}\omega+2}\frac{1}{\mathrm{j}\omega+5}=\frac{\mathrm{e}^{\mathrm{j}2\omega+4}}{(\mathrm{j}\omega)^{2}+7\mathrm{j}\omega+10}$$

Properties of the continuous time Fourier transform

11.Frequency domain convolution property

If $x_1(t) \xleftrightarrow{\mathscr{F}} X_1(\mathrm{j}\omega)$ $\quad x_2(t) \xleftrightarrow{\mathscr{F}} X_2(\mathrm{j}\omega)$

then $$x_1(t) \cdot x_2(t) \xleftrightarrow{\mathscr{F}} \frac{1}{2\pi}[X_1(\mathrm{j}\omega) * X_2(\mathrm{j}\omega)]$$

Properties of the continuous time Fourier transform

Energy spectrum of a non-periodic signal $x(t)$

$$W=\int_{-\infty}^{\infty}|x(t)|^2\,\mathrm{d}t$$

$|X(\mathrm{j}\omega)|^2$ ----Energy spectrum

$$=\int_{-\infty}^{\infty}x(t)[\frac{1}{2\pi}\int_{-\infty}^{\infty}X^*(\mathrm{j}\omega)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}\omega]\mathrm{d}t$$

$$=\frac{1}{2\pi}\int_{-\infty}^{\infty}X^*(\mathrm{j}\omega)[\int_{-\infty}^{\infty}x(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t]\,\mathrm{d}\omega$$

$$=\frac{1}{2\pi}\int_{-\infty}^{\infty}X^*(\mathrm{j}\omega)\cdot X(\mathrm{j}\omega)\mathrm{d}\omega=\frac{1}{2\pi}\int_{-\infty}^{\infty}|X(\mathrm{j}\omega)|^2\,\mathrm{d}\omega$$

12.Theorem of conservation of energy

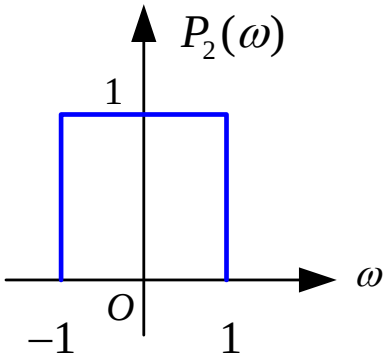
(Parseval Theorem of Conservation of Energy)

$$\int_{-\infty}^{\infty} |x(t)|^2 \mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\mathrm{j}\omega)|^2 \, \mathrm{d}\omega$$

The normalized energy of the aperiodic energy signal in the time domain is equal to its normalized energy in the frequency domain, and **the energy is conserved**.

Properties of the continuous-time Fourier transform

[Example] Calculate $\int_{-\infty}^{\infty} (\frac{\sin t}{t})^2 \mathrm{d}t$



Solution:

$$\mathscr{F}\{\frac{\sin t}{t}\} = \pi p_2(\omega)$$

According to Parseval Theorem of Conservation of Energy, it can be obtained that

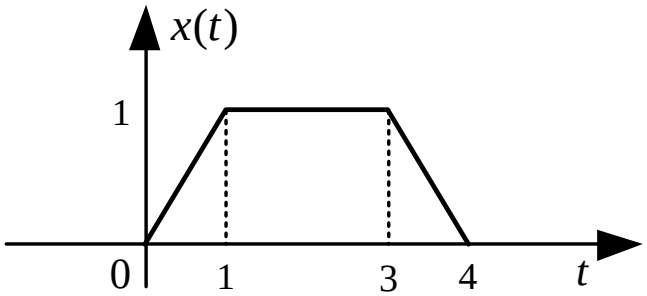
$$\int_{-\infty}^{\infty} (\frac{\sin t}{t})^2 \mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\pi p_2(\omega)|^2 \mathrm{d}\omega = \frac{1}{2\pi}\int_{-1}^{1} \pi^2 \mathrm{d}\omega = \pi$$

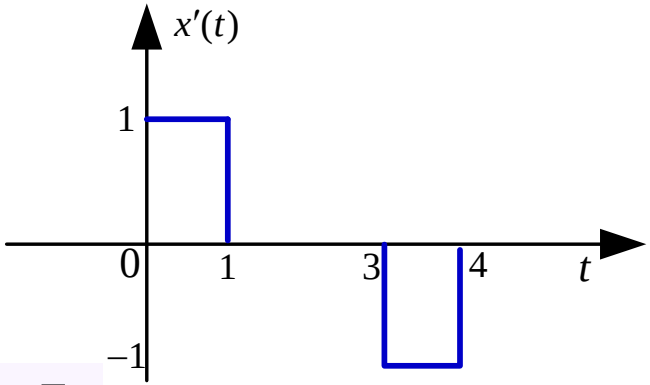
$$\int_{-\infty}^{\infty} (\frac{\sin 2t}{2t})^2 \mathrm{d}t = ?$$

$$\int_{-\infty}^{\infty} (\frac{\sin 4t}{4t})^2 \mathrm{d}t = ?$$

Examples of frequency domain analysis of continuous aperiodic signals

[Ex.] A known continuous-time signal $x(t)$ is shown in the figure, find its spectrum.





Solution: Method 1

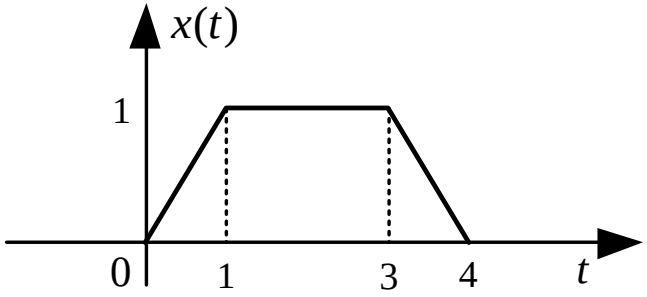
$$x'(t) = p_1(t-\frac{1}{2}) - p_1(t-\frac{7}{2})$$

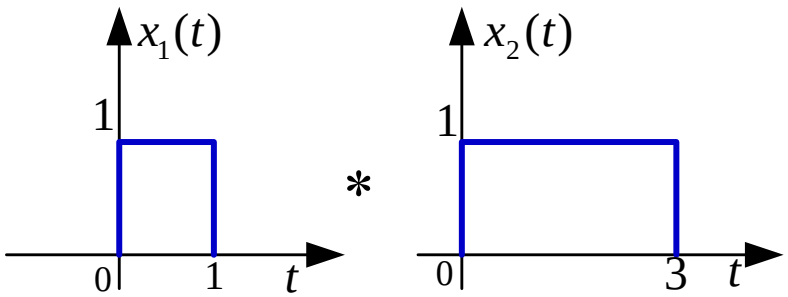
Based on the differential property in the time domain of Fourier transform, it can be obtained that

$$\mathrm{j}\omega X(\mathrm{j}\omega) = \mathrm{Sa}(\frac{\omega}{2})\mathrm{e}^{-\mathrm{j}\frac{\omega}{2}} - \mathrm{Sa}(\frac{\omega}{2})\mathrm{e}^{-\mathrm{j}\frac{7\omega}{2}}$$

$$X(\mathrm{j}\omega) = 3\mathrm{Sa}(\frac{\omega}{2})\cdot\mathrm{Sa}(\frac{3\omega}{2})\mathrm{e}^{-\mathrm{j}2\omega}$$

Examples of frequency domain analysis of continuous aperiodic signals





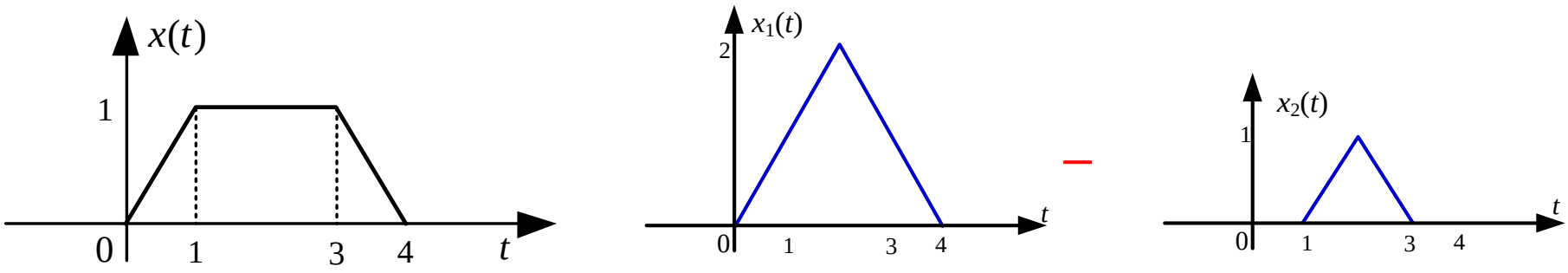
Solution: Method 2 $\quad x(t) = x_1(t) * x_2(t)$

$$X_1(\mathrm{j}\omega) = \mathrm{Sa}(\frac{\omega}{2})\mathrm{e}^{-\mathrm{j}\frac{\omega}{2}} \qquad X_2(\mathrm{j}\omega) = 3\mathrm{Sa}(\frac{3\omega}{2})\mathrm{e}^{-\mathrm{j}\frac{3\omega}{2}}$$

According to the convolution property of the Fourier transform, we can get

$$X(\mathrm{j}\omega) = X_1(\mathrm{j}\omega)X_2(\mathrm{j}\omega) \quad = 3\mathrm{Sa}(\frac{\omega}{2})\cdot\mathrm{Sa}(\frac{3\omega}{2})\mathrm{e}^{-\mathrm{j}2\omega}$$

Examples of frequency domain analysis of continuous aperiodic signals



Solution:Method III

$$x(t) = x_1(t) - x_2(t)$$

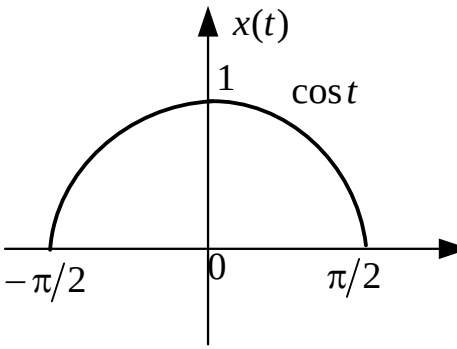
$$X_1(\mathrm{j}\omega) = 4\mathrm{Sa}^2(\omega)\mathrm{e}^{-\mathrm{j}2\omega} \qquad X_2(\mathrm{j}\omega) = \mathrm{Sa}^2(\frac{\omega}{2})\mathrm{e}^{-\mathrm{j}2\omega}$$

According to the linear characterization of the Fourier transform, it can be obtained as

$$X(\mathrm{j}\omega) = X_1(\mathrm{j}\omega) - X_2(\mathrm{j}\omega) = 3\mathrm{Sa}(\frac{\omega}{2}) \cdot \mathrm{Sa}(\frac{3\omega}{2})\mathrm{e}^{-\mathrm{j}2\omega}$$

Examples of frequency domain analysis of continuous aperiodic signals

[Ex.] A known continuous-time signal $x(t)$ is shown in the figure, find its spectrum.



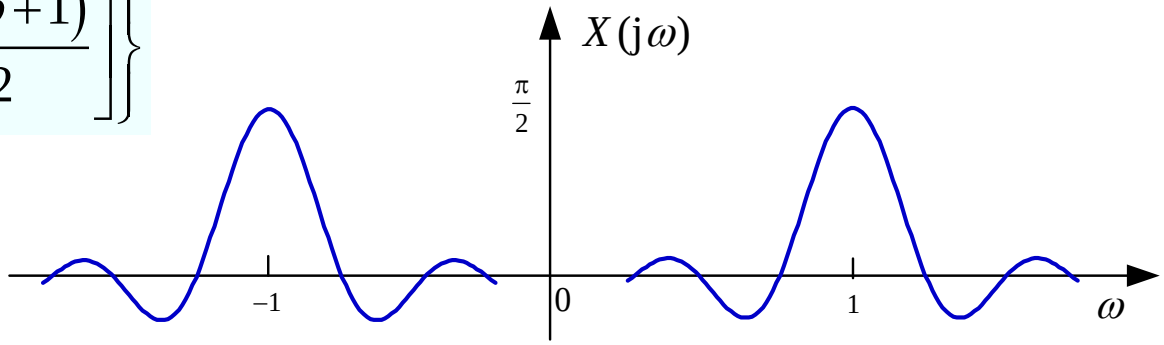
Solution: Method I

$$x(t) = \cos t \cdot p_{\pi}(t)$$

Based on the product property of the Fourier transform, it can be obtained

$$X(\mathrm{j}\omega) = \frac{1}{2\pi} \cdot \pi\left[\delta(\omega-1) + \delta(\omega+1)\right] * \pi \mathrm{Sa}\left(\frac{\pi}{2}\omega\right)$$

$$= \frac{\pi}{2} \cdot \left\{ \mathrm{Sa}\left[\frac{\pi(\omega-1)}{2}\right] + \mathrm{Sa}\left[\frac{\pi(\omega+1)}{2}\right] \right\}$$

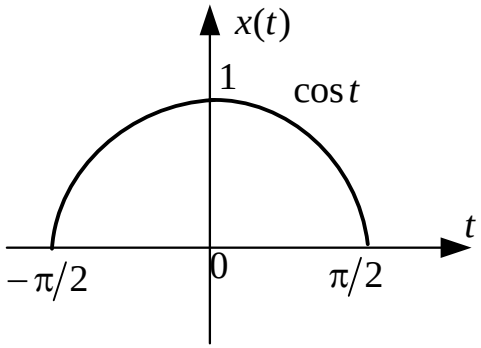


Examples of frequency domain analysis of continuous aperiodic signals

[Ex.] A known continuous time signal $x(t)$ is shown in the figure, find its spectrum.

Solution: Method II

$$x(t) = \cos t \cdot p_{\pi}(t)$$



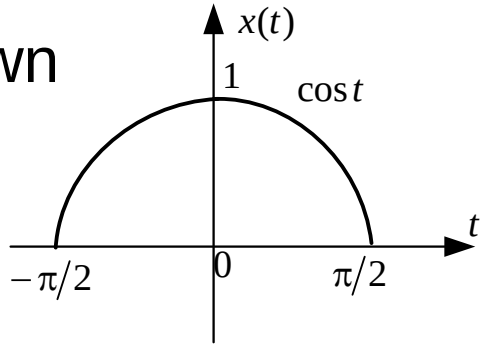
$$\mathscr{F}[p_{\pi}(t)] = \pi \mathrm{Sa}\left(\frac{\pi}{2}\omega\right)$$

According to the frequency shift property of Fourier transform, we can get

$$X(\mathrm{j}\omega) = \frac{\pi}{2}\left\{\mathrm{Sa}\left[\frac{\pi}{2}(\omega-1)\right] + \mathrm{Sa}\left[\frac{\pi}{2}(\omega+1)\right]\right\}$$

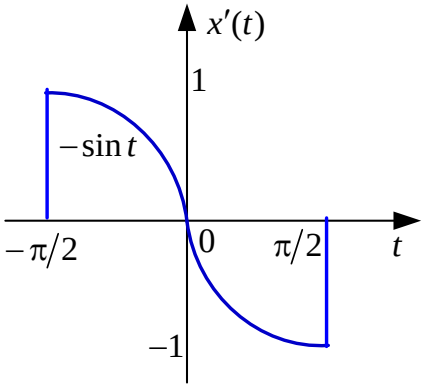
Examples of frequency domain analysis of continuous aperiodic signals

[Ex.] A known continuous time signal $x(t)$ is shown in the figure, find its spectrum.



Solution: Method III

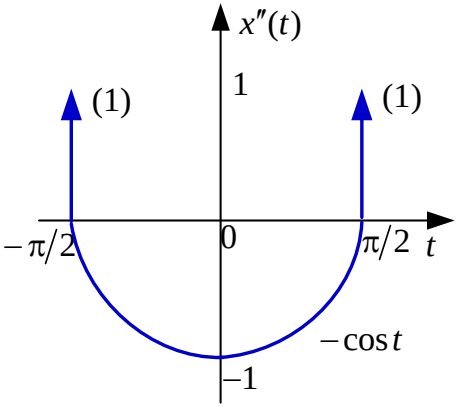
$$x''(t)=\delta\left(t+\frac{\pi}{2}\right)+\delta\left(t-\frac{\pi}{2}\right)-x(t)$$



According to the differential property of the Fourier transform, it can be obtained that

$$(\mathrm{j}\omega)^2 X(\mathrm{j}\omega)=\mathrm{e}^{\mathrm{j}\frac{\pi}{2}\omega}+\mathrm{e}^{-\mathrm{j}\frac{\pi}{2}\omega}-X(\mathrm{j}\omega)$$

$$X(\mathrm{j}\omega)=\begin{cases}\dfrac{2\cos\left(\dfrac{\pi}{2}\omega\right)}{1-\omega^2} & \omega\neq\pm1\\[2ex] \dfrac{\pi}{2} & \omega=\pm1\end{cases}$$



Summary of frequency domain analysis of continuous aperiodic signals

- Important Concept: Spectrum of a non-periodic signal

1) The representation of a non-periodic signal in frequency domain

2) Physical significance of spectrum of a non periodic signal

3) Methods of analyzing the spectrum of a non-periodic signal: The spectrum and Fourier transform properties of common basic signals are applied

- Mathematical tools used to analyze problems: Fourier transform

- Engineering applications: modulation, demodulation, frequency division multiplexing

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!